

LESSON NO. 66

Virus name The name “virus” has been derived from a Latin word meaning “slimy liquid” or “poison.” Viruses are microscopic parasites that lack the capacity to thrive and reproduce outside of a host body. These can multiply only in living cells of animals, plants, or bacteria. Latin word meaning poison or venom given by Beijerinck (1897).

WHAT ARE VIRUSES? Viruses are submicroscopic, obligate intracellular parasites. Most are too small to be seen by optical microscopes, and they have no choice but to replicate inside host cells. This simple but useful definition goes a long way toward describing viruses and differentiating them from all other types of organism. However, this short definition is not completely adequate. Viruses are unique in nature. They are the smallest of all self-replicating organisms, historically characterized by their ability to pass through filters that retain even the smallest bacteria. In their most basic form, viruses consist solely of a small segment of nucleic acid encased in a simple protein shell. Viruses have no metabolism of their own but rather are obliged to invade cells and parasitize subcellular machinery, subverting it to their own purposes.

Virus Size ▪ Submicroscopic Smaller than bacteria, Smaller than the smallest cell

LESSON NO.67 Importance of virology UNIQUE PROPERTIES ▪ Living as well as non-living characteristics ▪

Unable to cultivate in laboratory easily ▪ Simplest structure ▪ Complex interaction with host ▪ Replication within host

To study an important class of pathogens Many types of pathogens cause disease in humans. The most familiar are viruses and bacteria. Viruses cause diseases ranging from AIDS and smallpox to the common cold. They are essentially fragments of nucleic acid (DNA or RNA) instructions, wrapped in a protective shell of proteins and (in some cases) membrane. They use the basic transcription and translation machinery of their host cells for their replication.

Of all the bacteria we encounter in our lives, only a small minority are dedicated pathogens. Much larger and more complex than viruses, bacteria are usually free-living cells, which perform most of their basic metabolic functions themselves, relying on the host primarily for nutrition.

Some other infectious agents are eukaryotic organisms. These range from single-celled fungi and protozoa, through large complex metazoa such as parasitic worms.

Some rare neurodegenerative diseases, including mad cow disease, are caused by an unusual type of infectious particle called a prion, which is made only of protein. Although the prion contains no genome, it can nevertheless replicate and kill the host.

Even within each class of pathogen, there is striking diversity. Viruses vary tremendously in their size, shape, and content, and the same is true for the other pathogens. The ability to cause disease (pathogenesis) is a lifestyle choice, not a legacy shared only among close relatives

Virus life cycle Bacteria, fungi, and eukaryotic parasites are cells themselves. Even when they are obligate parasites, they use their own machinery for DNA replication, transcription, and translation, and they provide their own sources of metabolic energy. Viruses, by contrast, are the ultimate hitchhikers, carrying little more than information in the form of nucleic acid. The information is largely replicated, packaged, and preserved by the host cells. Viruses have a small genome, made up of a single nucleic acid type—either DNA or RNA. The genome is packaged in a protein coat.

Viruses replicate in various ways. In general, replication involves (1) disassembly of the infectious virus particle, (2) replication of the viral genome, (3) synthesis of the viral proteins by the host cell translation machinery, and (4) reassembly of these components into progeny virus particles.

To study viral cultivation in lab ▪ Tissue culture” ▪ Embryonated egg ▪ Live animal

LESSON NO.68 General Properties of VIRUS A virus is a sub-microscopic particle that can infect living cells.

Viruses are much smaller than prokaryotes, ranging in size from about 20–300 nanometers (nm), though some can be larger. Prokaryotes are typically 0.5–5.0 micrometers (µm) in length. For example, if a virus was about the size of three soccer balls lying side-by-side, then a prokaryote would be about the size of soccer field. Virus Size Foot and Mouth Disease Virus 27 nm Pox Virus 300 nm Structure An individual virus is called a virion. It is a tiny particle much smaller than a prokaryotic cell. Because viruses do not consist of cells, they also lack cell membranes, cytoplasm, ribosomes, and other cell organelles. Without these structures, they are unable to make proteins or even reproduce on their own. Instead, they must depend on a host cell to synthesize their proteins and to make copies of themselves. Although viruses are not classified as living things, they share two important traits with living things. They have genetic material, and they can evolve.

Simple Viruses ▪ a protective protein coat ▪ a nucleic acid genome Deoxyribonucleic Acid (DNA) or Ribonucleic Acid (RNA) ”

WHY VIRUSES NEED HOST? ▪ No enzymes of their own ▪ No metabolism ▪ Require host machinery

COMPLEX VIRUSES Simple viruses only contain protein coat and nucleic acid genome. Complex viruses also contains some additional contents e.g. ▪ Lipid ▪ Polysaccharides ▪ Trace elements

Lesson no.69 Structure of enveloped and non-enveloped viruses. Non-enveloped viruses are composed of capsid protein and nucleic acid (DNA or RNA), viz. nucleocapsid. Whereas enveloped viruses are composed of an envelope and nucleocapsid. Some viruses may have spikes to help attach to the host cell. Most viruses infect only specific host cells.

MORPHOLOGY ▪ Viruses come in a variety of shapes. ▪ Helical Viruses: Protein subunits interact with each other and with the nucleic acid to form a coiled, ribbon like structure: e.g. tobacco mosaic virus, influenza virus, rabies virus. Ebola virus. Structures with helical symmetry. ▪ Polyhedral shapes: The capsid shell is made of repeating subunits of viral protein (may be one kind of subunit or several, according to the virus). - All faces of the icosahedron are identical. like the influenza virus ▪ Complex shapes: Most viruses have either icosahedral or helical capsids, but some viruses do not fit into either category. Poxviruses and large bacteriophages are two important examples

Lesson no.70 Bacteriophages Introduction Even bacteria can get a virus! The viruses that infect bacteria are called bacteriophages, and certain bacteriophages have been studied in detail in the lab (making them some of the viruses we understand best). We'll take a look at two different cycles that bacteriophages may use to infect their bacterial hosts: ▪ The lytic cycle: The phage infects a bacterium, hijacks the bacterium to make lots of phages, and then kills the cell by making it explode (lyse). ▪ The lysogenic cycle: The phage infects a bacterium and inserts its DNA into the bacterial chromosome, allowing the phage DNA (now called a prophage) to be copied and passed on along with the cell's own DNA.

A bacteriophage is a virus that infects bacteria A bacteriophage, or phage for short, is a virus that infects bacteria. Like other types of viruses, bacteriophages vary a lot in their shape and genetic material. ▪ Phage genomes can consist of either DNA or RNA, and can contain as few as four genes or as many as several hundred. ▪ The capsid of a bacteriophage can be icosahedral, filamentous, or head-tail in shape. The head-tail structure seems to be unique to phages and their close relatives (and is not found in eukaryotic viruses).

Bacteriophage infections Bacteriophages, just like other viruses, must infect a host cell in order to reproduce. The steps that make up the infection process are collectively called the lifecycle of the phage. Some phages can only reproduce via a lytic lifecycle, in which they burst and kill their host cells. Other phages can alternate between a lytic lifecycle and a lysogenic lifecycle, in which they don't kill the host cell (and are instead copied

along with the host DNA each time the cell divides). Let's take closer look at these two cycles. As an example, we'll use a phage called lambda (λ), which infects E. coli bacteria and can switch between the lytic and lysogenic cycles.

LESSON NO.71 BACTERIOPHAGES: Properties A bacteriophage also known informally as a phage is a virus that infects and replicates within bacteria and archaea.

General Characteristics Virus architecture ,Virus particle called virion , Consists of nucleic acid surrounded by protein coat □ › Shapes Capsid › Isometric › Helical › Complex › Two types of virion › Naked – without envelope › Enveloped – surrounded by lipid membrane SIZE:

LESSON NO.72 MICROBES They are also minute, unicellular organisms that called as germs or microorganisms. They are too small to be seen with the unaided eye. Therefore, a microscope is used to observe them. They are found all around us. They live in water, soil, and in the air. The human body is home to millions of these microbes too, also called microorganisms. Some microbes make us sick; others are important for our health. The most common types are bacteria, viruses and fungi. There are also microbes called protozoa. These are tiny living things that are responsible for diseases such as toxoplasmosis and malaria.

MICROBIOLOGY It is the study of microorganisms. The field is concerned with the structure, function, and classification of such organisms and with ways of both exploiting and controlling their activities.

CLASSIFICATION OF MICROBES Classification of microorganisms has been a challenge. In the currently accepted scientific classification of Life, there are three domains of microorganisms: ▪ Eukaryotes ▪ Bacteria ▪ Archaea

LESSON NO.73 Classification of Microbes Introduction There is a need to classify microbes in order to study them conveniently, to identify them easily, to establish relationship between organisms and to establish the evolutionary relationships.

Bases of Classification: ▪ Similarities among them ▪ Morphology used to be the main feature prior to the advent of molecular biology ▪ Biochemical similarities ▪ Genetic similarities ▪ Evolutionary relationships

Five Kingdom Classification of Organisms The first person to divide living things into five broad kingdoms was North American ecologist Robert Whittaker. This researcher proved in 1959 that fungi were not plant organisms - previously it was thought that they were - and a decade later he proposed the creation of the fungi kingdom to differentiate them from plants. Whittaker's theory was widely accepted, and the scientific community thereby added a new group to the previous four-kingdom system, established by the American biologist Herbert Copeland in 1956.

Animal Kingdom The kingdom Animalia is the most evolved and is divided into two large groups - vertebrates and invertebrates. These animals are multi-celled, heterotrophic eukaryotes with aerobic respiration, sexual reproduction and the ability to move. This kingdom is one of the most diverse and comprises mammals, fish, birds, reptiles, amphibians, insects, molluscs and annelids, among others.

Plant Kingdom Trees, plants and other species of vegetation make up part of the Plantae kingdom - one of the oldest, and characterized by its immobile, multicellular and eukaryotic nature. These autotrophic things, whose cells contain cellulose and chlorophyll are essential for life on Earth since they release oxygen through photosynthesis. As regards their method of reproduction, this may be either sexual or asexual.

Fungi Kingdom This name is used to designate the fungi kingdom which includes yeasts, moulds and all species of mushrooms and toadstools. These multicellular aerobic heterotrophic eukaryotes have chitin in their cell walls, feed off other living things, and reproduce through spores.

Protista Kingdom This group is the most primitive of the eukaryotics and all the others are descendants of it. The Protista kingdom is paraphyletic - it contains the common ancestor but not all its descendants - and it

includes those eukaryotic organisms that are not deemed to be animals, plants or fungi such as protozoa. As it is so heterogeneous it is difficult to categorise it, since its members have very little in common.

Monera Kingdom This is the kingdom of microscopic living things and groups together the prokaryotes (archaea and bacteria). This group is present in all habitats and is made up of single-cell things with no defined nucleus. Most bacteria are aerobic and heterotrophic, while the archaea are usually anaerobic and their metabolism is chemosynthetic. The classification of the five kingdoms of nature remains the most accepted today, although the latest advances in genetic research have suggested new revisions and reopened the debate among experts. Such is the case for the sixth kingdom of Carl Woese and George Fox, who in 1977 divided bacteria into two types (Archaea and Bacteria), and the seventh kingdom of Cavalier Smith, who added a new group to the previous six for algae called Chromista.

Three Domain Classification It was proposed by Carl Woese in 1977. According to this system, organisms are divided into following domains.

1. Bacteria: These are prokaryotes that are usually single-celled organisms. Their cell wall is made up of peptidoglycan.

2. Archea: These are prokaryotes. Their cell wall has no peptidoglycan but has special lipids. They also vary in rRNA sequencing and have capability to live in extreme conditions. They include thermophiles and halophiles

3. Eukarya: They include the following kingdoms: o Protists (slime molds, protozoa, algae o Algae contain cellulose cell wall and are photosynthetic in nature, as they can produce oxygen. o Protozoans are motile and unicellular organism. o Slime molds are Protozoa like at one stage and fungus like in another. Fungi o Unicellular form is called yeast. o Multicellular form is known as mold or mushroom Plants Animals

Other Microbes Viruses: These are acellular organisms that consist of proteins and nucleic acids Viroids: These are composed only of circular ssRNA, for example, potato spindle tuber viroid Virusoids: These contain circular ssRNA and need helper viruses for their replication and encapsidation. These are also called satellite viruses Prions: These are defined as infectious protein particles.

Naming Organisms: Binomial Nomenclature Carl Linnaeus established the system of scientific nomenclature in 1735. According to this system, each organism has two names: the genus and specific epithet (descriptive word)

Scientific Names: Scientific names of microorganisms are italicized or underlined. The genus is capitalized, and the specific epithet is lowercase. Scientific names are “Latinized” and used worldwide. These names may be descriptive or honor a scientist. For example, the scientific name of *Escherichia coli* honors the discoverer, Theodor Escherich and describes the bacterium’s habitat—the large intestine, or colon. The name of *Staphylococcus aureus*, describes the clustered (staphylo-) spherical (cocci) cells and the gold-colored (aureus) colonies. After the first use, scientific names may be abbreviated with the first letter of the genus and the specific epithet: *Escherichia coli* and *Staphylococcus aureus* are found in the human body. *E. coli* is found in the large intestine, and *S. aureus* is on skin.

LESSON NO.74 Microscopy INTRODUCTION An unaided human eye can only distinguish two objects only if these are 0.2 mm apart from each other. Thus, the resolving power of the human eye is 0.2mm or 200µm. On the other hand, objects to be studied by biologists are too small to be seen with naked eye. For instance, most cells have a diameter of about 10µm only. For observing such tiny objects, scientists make use of microscopes.

MICROSCOPE A microscope is an instrument that magnifies objects otherwise too small to be seen, producing an image in which the object appears larger. It was first invented by Antonie Van Leeuwenhoek. Thus, he was called as the “Father of Microbiology”. Most photographs of cells are taken using a microscope, and these pictures can also be called micrographs.

HISTORY Microscope was firstly invented by Antonie van Leeuwenhoek. For the same reason he is known as the Father of Microbiology

USES OF MICROSCOPE ▪ Microscopes are useful in ▪ Diagnostics ▪ Identification of organisms ▪ Histology – Tissue analysis ▪ Examining Forensic Evidences ▪ Studying cells and Micro-organisms ▪ Scientific Research

PARTS OF COMPOUND MICROSCOPE **Eyepiece** or ocular lens having 10x magnification power, using which you can observe the magnified image of object. **Arm** of microscope, which connects eyepiece to base of microscope. **Stage** where we place our specimen slide to be observed. **Stage clips** to hold the slide in proper place during observation, there are **stage clips** which fix the slide in its particular position. **Objective lenses** of 4X, 10X, 40X, 100X magnification powers that are mounted on a rotating turret called as nosepiece. Nosepiece can be rotated to change the lens, which clicks into position when the lens is set. The four lenses are color coded as well to give an idea of their magnification power. o The lowest power objective lens is coded red and has a magnification of 4x. o The lens coded yellow has a magnification power of 10x. o The high-power magnification lenses, coded blue and green respectively, have a power of 40X and 100x. However, 100x is also called as oil immersion lens and its reason will be explained shortly. **Condenser**: Whose function is to focus the light rays onto the specimen under observation. **Diaphragm** along with its control lever. This is used to control the size of aperture through which light passes. Alternately the intensity of light can also be controlled through another knob known as the illumination control knob. **Light source/Illuminator** from where the light emits. Light is then passed to the condenser lens to be focused on the specimen. **Coarse adjustment knob** that moves the stage up and down and for focusing the specimen. **Fine adjustment knob**: As you know every person have specific vision power so while observing every individual focus the specimen at different position. The Fine adjustment knob here is used to deal with such differences and helps to focus the object without moving the specimen or slide. This in simple terms is used for someone who otherwise wears glasses. **Base** provides basal support to the microscope..

LESSON NO.75 MAGNIFICATION POWER Magnification is the ability of lens to enlarge the object.

Magnification power of microscope is determined by the focal lengths of the two lens systems. To get the total magnification take the power of the objective lens used that may be 4X or 10X or 40x or 100X and multiply it by the power of the eyepiece that is 10X. For example, if you are using objective lens of power 40X, then multiply 40 with 10, because power of eyepiece is always 10x. As a result, total magnification of microscope will be 400X.

REFRACTIVE INDEX It is defined as the ability of a medium to bend the light when passing from one medium into another. Light rays move in a straight line through a single medium

RESOLUTION POWER It is the Ability of the lenses to distinguish between two closely lying objects as separate. Light microscope resolving power is 0.2 μm .

LESSON NO.76 Compound Microscope In a compound microscope, magnifying powers of two convex lenses, eyepiece and objective lens, are used to produce a magnified image of very small objects. Light from a light source pass through a thin object. The objective lens produces a magnified 'real image' of object. This image is again magnified by the ocular lens eyepiece to obtain a magnified 'virtual image' that is a final image and can be seen by eye through the eyepiece. Thus, the light passes directly from the source to the eye through the two lenses, the field of vision is brightly illuminated. That is why; it is a bright-field microscope. Transparent specimens can be stained before their observation. These stains usually color different parts of the cells.

Dark Field Microscope- It is used for the Examination of unstained microbes. It contains special kind of condenser, with an opaque disc obstructing the path of light from the light source centrally but allowing a peripheral ring of light to pass. As a result, bright specimen can be viewed against a dark background.

Phase Contrast Microscope- that is applicable in Cell culture. It is used to observe living cells in their natural state without any previous fixation or labeling. Its principle is such that, when light passes through cells, small phase shifts occur. These phase shifts are converted into changes in amplitude, which can be observed as differences in image contrast

Fluorescent Microscope – is another variation of microscopes in which specimens are formerly stained with Florescent dyes, for example DAPI(4',6-diamidino-2-phenylindole) and Hoechst stains. During microscopy, specimen is illuminated with light of a specific wavelength that is absorbed by fluorophores and converted into a light of longer wavelength. As a result, fluorescent specimen can be easily observed.

LESSON NO.77 Electron Microscope: It has high magnification and resolution. It uses a beam of accelerated electrons as a source of illumination. There are two main types of electron microscope including: o Transmission Electron Microscope (TEM) o Scanning Electron Microscope (SEM) Table 77.1 Main differences between a SEM and a TEM

Lesson no.78 Digestive System Nutrition Acquiring energy from environment is one essential property of life. Plants acquire energy from sunlight and prepare their food. We call them producers. Animals, however, acquire energy from either plants or other animals or organic materials. We call the consumers or decomposers.

Common Modes of Nutrition in Animals ▪ Herbivores o Plant eaters, e.g., goat ▪ Carnivores o Flesh, meat eaters, e.g., lion ▪ Omnivores o Mixed food, e.g., bear, crows ▪ Detritivores o Dead organic matter, e.g., earthworms. Dentition is according to the mode of nutrition

Mouth Parts also are also according to the mode of nutrition Animals have different types of tongues according to their mode of nutrition. For example, frogs have forked, inverted and sticky tongue to catch insects. Tongue of chameleon is also sticky and very long. Beaks of birds make another example. Birds have a variety of beaks; long, small, curves and many other. All of these are present in birds according to their mode of nutrition for example parrots have long curves beak for cutting and breaking nuts. Pelican have a very long and wide for catching fish from water. Snouts in mammals make another example. Mammals have different snouts according to their food.

LESSON NO.79 Nutrition, Ingestion, Digestion, Absorption, Assimilation, Elimination ▪ After acquiring food and ingestion, animals have to digest it. ▪ Digestion is actually, breaking down food into small ingredients. For example, carbohydrates to glucose. ▪ Absorption is movement of food into blood after digestion. ▪ Assimilation is utilizing those broken small molecules for getting energy for organism's functions. ▪ Elimination is removal of the undigested matter from the body.

Nutrition and Digestion in Human beings Humans are omnivores, i.e., we eat plants (vegetables) and meat (chicken, beef). Digestive system of humans hence is adapted accordingly.

LESSON NO.80 Basic Components of Human Digestive System ▪ Alimentary canal o From mouth to anus ▪ Alimentary canal in humans consist of following parts: o Oral cavity o Pharynx o Esophagus o Stomach o Intestine (small and large) Rectum and anus ▪ Accessory glands o Liver, Pancreas

Sensing the Food We look at the food and smell when we get a food. These are called sensory qualities of food. This is the first part of food selection and foods which are improper are rejected. If the food looks inappropriate or it smells bad, it is rejected. This is a human adaptation just like other animals.

Alimentary Canal – From mouth to anus

The Oral Cavity Food selection is one function of the oral cavity. Oral cavity has tongue with taste buds. Food selection takes place by taste buds if food tastes bad it is rejected. Tongue converts the ground food mass to a bolus. Grinding the food is the other important function of oral cavity. Oral cavity contains teeth for this

purpose. Dentition is according to the food types. Teeth are of four types: incisors, canine, premolars and molars. Humans have all four types of teeth; few animals have others. Lubrication and chemical digestion is another function of the oral cavity. Salivary glands secrete saliva which contains amylase for digestion of starch. It also contains mucous for lubrication of food.

Pharynx Pharynx is a part of digestive system. It lies just behind the mouth and nasal cavity and above the esophagus and larynx. Food bolus passes through pharynx to enter the esophagus. Pharynx helps swallowing by closing the trachea and nasal cavity pathways.

Esophagus It is a long tube starts from pharynx and enters stomach. From pharynx, food enters esophagus. In esophagus, food moves down to stomach by a series of muscular contractions called “peristalsis”. Bolus moves down by alternate contraction and relaxation of muscles of esophagus. Sometimes an antiperistalsis occur, i.e., movement from stomach to mouth that result in vomiting.

Stomach Stomach is mainly a storage organ. It also helps in digestion of proteins. Stomach has guarded openings on either side. There is a sphincter (group of muscles) which is pyloric sphincter that controls movement of food from stomach to intestine. The other sphincter is present where esophagus meets stomach. Gastric glands in stomach produces gastric juice, which contain mucus, hydrochloric acid and an enzyme called pepsinogen. Pepsinogen is converted into pepsin which is its activated form. Pepsin converts proteins to peptides. Protection of the stomach walls from the acid particularly is important, mucus secreted by the stomach walls cover the stomach lining to protect it from acidic damage. Stomach is also designed to mix up the food with enzymes and acid. Churning movements of stomach walls thoroughly mixes the food with gastric juice. Churning converts food to chyme, small part of which enters the small intestine by pyloric sphincter

LESSON NO.81 Small Intestine It consists of three parts: duodenum, jejunum and ileum. Duodenum (first 25 cm of intestine) is the first part where most of the digestion occurs. Bile enters here from liver and helps emulsification and digestion of lipids. Pancreatic juice from pancreas also enters here, which contains trypsin, pancreatic amylase and lipase for digestion of protein, carbohydrates and lipids. Intestinal juice is also released, which help in digestion of all foods. Jejunum is the next 2.4 m of the small intestine; remaining digestion takes place here. Ileum is the last 3.5 m long part where absorption takes place. Its lumen is highly folded. It have extensions of the epithelium called villi and microvilli, which increases the surface area to a greater extent to increase absorption.

LESSON NO.82 Large Intestine Undigested food comes in the large intestine. It consists of various parts including caecum, colon, and rectum. Absorption of water from the undigested matter occurs in colon. Water is returned to blood vascular system. Remaining semi solid mass is called feces that are are stored in rectum, when rectum is full it gives rise to a reflex for defecation and feces are excreted out. Intestinal Motility is an important factor. If the motility in large intestine is increased, then there is less water reabsorption and result is a situation called diarrhea. On the other hand, if motility is decreased then constipation is the result which means that feces will have less than normal water.

Liver and Pancreas Liver produces a secretion called bile which helps in fat digestion and emulsification. Bile is stored in gall bladder and is released by its contraction. Liver performs various functions called deamination, converting ammonia to urea, destroys old blood cells and manufactures fibrinogen, which is blood clotting factor.

LESSON NO.83 Transport System & Blood - The Circulatory Fluid **Why transport systems?** Organisms have to exchange materials with the environment. They also need to distribute nutrients, gases and metabolic products within their body.

Living organisms ▪ acquire energy or food from the environment; they have to distribute energy and food within the body. ▪ need an intake of oxygen and removal of carbon dioxide from their bodies. ▪ need to remove toxic materials and waste products from their bodies. ▪ have to distribute hormones and other materials in the body. ▪ have to exchange materials with the environment For all that, they require a Transport System.

Transport systems in organisms A transport system should fulfill the requirements of the organism for all of the above functions like nutrient distribution or gas exchange. We take some examples here: ▪ Unicellular organisms o e.g., amoeba ▪ Simple multicellular organisms o Small organisms, e.g., sponges, hydra ▪ Plants and animals

Unicellular organisms Unicellular organisms have a simple body, just one cell. Their requirements are also few. For example, amoeba performs the transport via cytoplasmic streaming which is the movement of their cytoplasm and few organelles. Food vacuoles distribute the food in the body. Contractile vacuoles remove the water from the body. Cytoskeleton helps in the movement of all the above parts.

Simple multicellular organisms Simple multicellular organisms transport materials by simple methods. Like hydra have a mouth and a body cavity; water enters through mouth and transport of materials takes place in the body cavity. Body wall also exchange materials with the water outside as organism is aquatic. Hence, water act as a medium of transport. Few cells help in transport, which are present inside the body towards body cavity.

Transport in plants occurs by roots and conducting tissues called phloem and xylem. Roots absorb water and materials from soil. Plants remove water through a process of transpiration. The leaf has guarded openings called stomata, which open and close to remove water.

Transport in animals Animals have a system called circulatory system for transport. There are two kinds of this system called an open circulatory system and closed circulatory system. In an open circulatory system, the circulatory fluid is freely moving in the body cavity bathing tissues and there is/are pumping heart/s but these only send blood to some vessels which are then connected to body cavity. In closed circulatory system, the blood never leaves vessels and heart and is always present inside the vessels.

LESSON NO.84 Basic components of human circulatory system Blood is the circulatory fluid which is the medium for transport. Blood vessels are the tubular system for transport of blood within the body. Heart is the pumping organ that pumps the blood toward the body.

Composition of blood Human blood consists of plasma (fluid) and cellular content (cells). Plasma is the fluid with dissolved and un-dissolved materials. It consists mainly of water, which have many inorganic and organic materials dissolved in it. Cellular content consists of Red Blood Cells (RBCs), White Blood Cells (WBCs) and Platelets. Red blood cells are also called erythrocytes and white blood cells are also called leukocytes. Plasma consists of 55% of the blood. Cells constitute 45% of the blood Average human body has about 5liters of blood. Blood contents could be separated based upon their densities and weights by various methods including centrifugation and sedimentation.

Functions of the main components of blood . Blood plasma Plasma consists of 55% of the blood, It consist of 92% water. 0.9% of it consists of salts – NaCl, Mg, Zn ions etc. The salts maintain the pH of the blood. Plasma has 7-9% part consist of proteins. Important proteins include antibodies, fibrinogen and albumin. Plasma also contains digested food, nitrogenous wastes, hormones, respiratory gases etc.

LESSON NO.86 Blood groups in human beings and blood transfusion Blood cells have some specific protein present on their surfaces called antigens, which are meant for the identification of the blood cells. About 29 blood groups are identified. The most important blood group systems are, however, two named as ABO and Rh blood group systems. There are four main types of blood groups present in human beings according to ABO

blood group system. These blood groups are based upon presence or absence of two antigens on surface of RBCs called antigen A and antigen B (Table 1). Other important blood group is based upon presence or absence of Rh antigens on the surface of RBCs. The people have Rh antigen on the surface of their RBCs are called Rh positive and those who do not have it are called Rh negative. ABO system works with the help of Rh system. So that we talk of the blood groups based upon ABO system with Rh positive or negative. For example, A +ve means that the person has an A blood type with Rh antigens present. Table 1

Blood transfusion Blood transfusion is required in case of some diseases, surgery or in case of injuries. Blood groups of donor and recipient should match, otherwise could cause agglutination reactions leading to even death of the patient. Blood groups are matched using antigen antibody reactions. A blood group O -ive is called Universal donor because it has not antigens, hence, it cannot react with any of the antibodies in recipient's blood. Universal acceptor is AB+ blood group which has all antigens.

LESSON NO.87 CIRCULATORY SYSTEM IN HUMANS – BLOOD VESSELS & HEART **Circulatory System in Humans**

Circulatory system in humans consists of blood, vessels and heart. Blood is the circulatory fluid. We can call it a liquid tissue. Blood consists of plasma – the watery fluid and cells – the cellular content. Blood has been described in detail in last lecture. Here we talk about heart and blood vessels. Heart is a pumping organ and is highly muscular. Function of the heart is to pump blood with pressure towards lungs and body. Vessels are of three main types; arteries, veins and capillaries. Arteries carry blood from the heart to body. Veins return blood from body to heart. Capillaries connect arteries to veins. Function of the capillaries is the exchange of materials.

Structure and functions of blood vessels Arteries and veins are elastic in their structure. These also have a layer of muscles. These consist of 3 layers: o External layer - connective tissues layer o Middle layer, consist of smooth muscles o Inner layer, consist of endothelium Lumen of both of these is meant for blood flow. Arteries have a lumen smaller in diameter and veins have large.

Arteries carry blood from heart to body. These face maximum pressure. Their function is to distribute blood to all the body tissues. Blood is pumped in the arteries by the heart with very high pressure that is why arteries are thick walled to withstand high pressure.

LESSON NO.88 Veins have to collect blood from the body and return it to the heart. Veins face less pressure, pressure almost negligible in major veins. Blood flow in major veins against gravitational pull and with low pressure so that veins have valves that prevent backflow of blood. In many parts of the body like arms and legs there are muscles that contract and relax to move the blood towards heart.

Capillaries have to exchange materials with tissues; these originate from arteries and pass from tissues then joins to make veins. Capillaries have to carry out exchange of materials with the tissues that is why their walls are just one celled thick. Capillaries when enters tissues makes branches, where they branch groups of muscles are present which may open or close to increase or decrease the blood flow towards tissues. Main arteries and veins in human arterial and venous systems ▪ head, shoulders, arms other body organs like kidneys, gonads, liver etc

LESSON NO.89 Structure of human heart Heart in human beings is a muscular pump to push blood towards body and lungs. Heart in humans is of the size of a clenched fist. Heart consists of four chambers including 2 atria and 2 ventricles. Atria receives blood from body and lungs. Ventricles push blood from heart to body and lungs. These chambers (atria and ventricles) are connected to each other. At their junction valves are present, which are responsible to prevent backflow of blood. Left side of the heart, particularly ventricle is stronger because it has to push blood to the whole body in comparison to right side.

Double pumping action of heart and role of valves \Deoxygenated blood from body enters the heart through major veins (vena cava) to right atrium. When right atrium contract, blood enters the right ventricle. There is a valve called tricuspid valve that prevents backflow of blood. On the other side, oxygenated blood from lungs enters the left atrium and when left atrium contract it enters the left ventricle. At the junction of left atrium and ventricle another valve is present called bicuspid valve that prevents backflow of blood. Tricuspid and bicuspid valves are collectively called atrioventricular valves.

In fact, both atria are filled at the same time and also contract at the same time to push blood in the ventricles. Both ventricles also contract at the same time to push blood toward body (left one) and lungs (right side). Right ventricle pushes blood in the pulmonary artery towards lungs. At this junction a valve is present called pulmonary valve that prevent backflow of blood from pulmonary arteries to right ventricle. Left ventricle pushes blood towards aorta, which is the main artery distributing blood the body. This junction is also guarded by a valve called aortic valve, which prevents backflow of heart from the aorta to left ventricle. Pulmonary and aortic valves are collectively called as semilunar valves.

Self-excitatory system of the heart Heart has self-excitatory system. Heart has a group of muscles called sinoatrial node which produces impulse (electrical activity) that spread in the heart to contract its various chambers systematically. Impulse spread toward atria first and causes the atria to contract. Then it enters another node called atrioventricular node where its magnitude is increased and then it enters the ventricles and the ventricles contract. The self-excitatory group of muscles of the heart is called pacemaker.

Cardiac Cycle Cardiac cycle is the time period from when the blood enters the heart to the time when blood is pushed by the heart. We can also say that it is the time period from atrial and ventricular diastole to ventricular systole. Cardiac cycle consists of following steps: ▪ Both atria and ventricles are relaxed. This is the time when both atria are filled with blood from body and lungs. All heart muscles are relaxed, which means that these are at diastole. ▪ Then both atria contract and blood are pushed into ventricles. This is called atrial systole. At this time, both AV valves are open and both semilunar valves are closed. Then the ventricles contract and push blood into aorta and pulmonary artery. This is called ventricular systole. With this, a cardiac cycle is complete. ▪ Then again, both atria and ventricles are relaxed, i.e., come back to the initial state.

LESSON NO.90 Double Circulation Human circulatory system is called a double circulatory system because heart receives and sends blood from and to lungs (pulmonary circuit) and body (systemic circuit). Look at the diagram below for a detailed view. ▪ Systemic circulation o From heart to body and vice versa o Head, shoulders and arms o Lower body organs ▪ Pulmonary circulation : From heart to lungs and vice versa.

LESSON NO.91 Why respiration? ▪ Living organisms needs energy for their activities. ▪ Energy is extracted by complex metabolic processes from food or photosynthesis. ▪ Respiration is the process of exchange of gases to produce energy in biological form.

What is respiration? Respiration consist of processes involves exchange of gases in living organisms or metabolic reaction to produce energy. Respiration is one of the most important metabolic activities. It is of two types: o Organismic respiration consists of breathing or ventilation, i.e., inhalation and exhalation movements and gas exchange. o Cellular respiration consists of metabolic reactions to produce energy by utilization of oxygen, production of carbon dioxide and extracting and conserving energy from food molecules in biological form, such as ATP.

Cellular respiration Cellular respiration is of two types: ▪ Aerobic, in which oxygen is utilized to produce energy. Example is electron transport chain in mitochondria. ▪ Anaerobic, in which oxygen is not involved but energy is produced using other molecules. Example is fermentation in some bacteria.

Organismic respiration Organismic respiration is the ventilation or breathing, i.e., the movements which causes inhalation and exhalation. Air contains gases and exchange of gases occurs in the specialized organs, for example, lungs in humans.

Difference between cellular and organismic respiration ▪ Cellular respiration is a series of events of metabolic reactions for production of energy. ▪ Organismic respiration is the process of coordinated movements of body that results in inhalation and exhalation of air. We also call it breathing or ventilation.

Respiration in air and water Respiration in air is easier in comparison to water because air is lighter, and water is denser. Also, air is much more saturated with oxygen than water. So that gas exchange in air is easier in the water than in water.

Respiration in plants Gas exchange in plants occurs by stomata in mesophyll cells. Stomata are openings guarded by guard cells; these are used for transpiration in plants through water vapors. Roots of land plants get the oxygen required by the plants from air present in soil. Aquatic plants acquire their oxygen from oxygen dissolved in water.

Respiration in animals In animals, respiration occurs by means of specialized organs, for example gills in fishes and lungs in terrestrial animals. The most important property of the respiratory organs is respiratory surfaces that support and facilitate the process of respiration. Properties of Respiratory surfaces

Properties of Respiratory surfaces include a large and moist surface area, a thin epithelium, ventilation (air availability) and a capillary network.

Respiration in some small animals In small animals, respiration occurs by means of general body surface or specialized systems. Hydra is an aquatic organism, which have a mouth and body cavity through which water moves in and out. Cells of the inner body wall exchange gases with that water. Cells of the epidermal layer exchange gases with the water. In some other organisms like earthworms which have a closed circulatory system the exchange of gases takes place by the capillaries, which are present in closed vicinity with the body wall. In the arthropods, especially insects like cockroach a very special system of gas exchange exist called the tracheal system. Tracheal system is a system of tubes which are extensions of the body wall and makes a network of tubes called tracheae, tracheoles and air sacs in the body. These have openings to the exterior called spiracles. Air enters in through spiracles and enters the trachea and tracheoles, which are present close to the tissues. Gas exchange is easier because in trachea only air is present, which is a very lighter medium. Tracheal system is a very efficient system of respiration.

Respiration in fish, frog and birds **Fishes** Fishes are aquatic animals; these have respiratory organs called gills. Water enters through mouth into the gills where gas exchange occurs and then water move out from gills. Heart in fishes is called a single circuit heart, which pumps blood towards gills where gas exchange occurs and then blood moves towards body.

Frogs Frogs are called a transition between aquatic and land animals. These live in water and land both. These have three different types of respiratory system at different stages of life. When a frog is hatched for an egg and becomes a larva, it lives in water and respire through gills. When it becomes an adult frog it respire by lungs and skin. Frog's skin has a network of blood capillaries so that gas exchange may occur there.

Birds Birds have a very efficient system of respiration. Birds respire through lungs. They also have air sacs in their in their body and bones, which also helps in capturing air and hence efficiency of gas exchange is improved.

Respiratory organs Land animals have their respiratory organs named lungs. In aquatic animals, gills are the organs for gas exchange.

LESSON NO.92 Respiratory system in humans Respiration in human beings occurs by lungs and associated air passageways. Human respiratory system consists of: Air passageways in humans ▪ Nostril and nasal cavity ▪ Pharynx ▪ Larynx ▪ Trachea ▪ Bronchi ▪ Bronchioles

Lungs in humans ▪ Pleural cavity ▪ Body of lungs ▪ Alveoli – the smallest units

Structure and functions of various parts of air passageways: Nose and nasal cavity Nose is the part for inhalation. Nose and nasal cavity are lined with ciliated epithelium and its surface is covered with mucus. The ciliated and moist surface traps dirt and other particles. The air, which enters inside is filtered, moist and warm.

Pharynx It is a small muscular passage, which is lined with mucous membrane. Air moves down to larynx through pharynx.

Larynx: the voice box Larynx is a complex cartilaginous structure, through which air moves down to trachea. Its opening is ciliated and covered with mucous. Mucus membrane is stretched across into thin edged fibrous bands called vocal cords. Vocal cords help in voice production.

Trachea (windpipe) Trachea is a long tube stretched from larynx to lungs. It has a tubular structure, and it lies ventral to the esophagus. It has C-shaped cartilaginous rings which are supporting structure and prevent tracheal collapse. Trachea extends through the chest cavity or thorax. In thorax, it divides into 2 tubes called bronchi one entering each lung.

Bronchi, bronchioles and alveoli Trachea divides to form bronchi in thorax region. Bronchi enter the lungs and divide further to make bronchioles (when they attain a diameter of 1mm or less). Bronchi have same cartilaginous rings but these become irregularly distributed plates when reaches at the end of bronchioles.

Alveoli Bronchioles keep dividing further and enter deep into the lungs to give rise to air sacs. Air sac is the functional unit of lungs. Air sacs consist of several microscopic structures called alveoli. Alveoli and air sacs are covered by a network of capillaries. This network is the site of gas exchange.

LESSON NO.93 Structure of lungs There is a pair of lungs, one is right and other is left. These are present in chest cavity. Lungs are protected by ribs and pleural cavity (double membrane fluid filled cavity). Lungs are spongy in structure because of the presence of millions of alveoli. Below lungs a muscular floor of the chest is present, which is called diaphragm.

Breathing Breathing is the process of inhalation and exhalation. Lungs are spongy, cannot pull or push air themselves. Breathing movements occurs with the help of pressure and movements of diaphragm. Diaphragm is dome-like structure present as the floor of chest cavity. When it is contracted, it becomes less dome-like, which results in expansion of chest and inhalation of air. When it is relaxed then it becomes more dome-like and pressure in the chest is decreased, which results in exhalation of air. When muscles between the ribs contract, the rib cage is elevated; and when relax, ribs settle down. This movement also helps in the process of breathing.

Exchange of gases in alveoli Gases are exchanged against a pressure difference in the blood and lungs (alveoli). Blood capillaries are distributed around alveoli in very thin layers. Blood cells and alveolar air are very close for exchange of gases. Gases are exchanged at the alveoli. Hemoglobin (Hb) in blood cells help in exchange of oxygen against the concentration difference. $Hb + O_2 \rightleftharpoons HbO_2$ (reversible if pressure of gases in air is changed)

Transport of CO₂ Carbon dioxide is more soluble and dissolves in tissue fluids. From here it enters into blood capillaries. Carbon dioxide is transported in blood as: • o 20 % as carboxyhemoglobin o 5% by few plasma proteins o About 70 % as bicarbonate ions combined with sodium. o $CO_2 + H_2O \xrightarrow{\text{Carbonic anhydrase}} H_2CO_3$ o $H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$

LESSON NO.94 Factors affecting gas exchange ▪ **Carbon dioxide** If CO₂ increases, then O₂ tension decreases. The result is that capacity of hemoglobin for O₂ is reduced. ▪ **Temperature** Rise in temperature decreases the oxygen carrying capacity of hemoglobin. ▪ **pH** If pH declines, then oxygen carrying capacity declines. ▪ **H⁺ ions** bind to hemoglobin to decrease oxygen carrying capacity. Lung capacity and effect of exercise on breathing rate In humans, fully inflated lungs have a capacity of 5 liters. Normally, at rest or sleep exchange is half a liter. During exercise, it may increase up to 3.5 liters. It means that there is a 1.5 L residual volume that cannot be expelled. Inhalation per minute is 15-20 times; during exercise it may be up to 30 times to fulfill the needs of exercising muscles.

LESSON NO.95 EXCRETORY SYSTEM Introduction and need of excretion Living organisms have to metabolize which consist of anabolism and catabolism. Anabolism consists of all reactions for formation of compounds. Catabolism is break down of various compounds including glucose, proteins and lipids for the formation of energy. Catabolism of carbohydrates, proteins and lipids carbohydrates + oxygen CO₂ + H₂O + energy proteins + oxygen CO₂ + H₂O + NH₃ + energy lipids + oxygen CO₂ + H₂O + energy

Metabolic wastes We can see in the reactions above that metabolic waste products are carbon dioxide, water and ammonia. These are waste and surplus materials. These have to be removed. Ammonia is highly toxic for the tissues. Water and CO₂ are dangerous for the body if these are present in excess. Rise in CO₂ may result in decrease in pH in blood which is dangerous. Ammonia is highly toxic and has to be converted into less toxic form. But those forms also need to be removed. Changes in water concentration in tissue fluids also create problems. Higher concentration may cause dropsy, which is accumulation of water in tissues. Decrease in water concentration may lead to dehydration. For all of these reasons an excretory system is required.

Excretion in Vertebrates: Major waste products in vertebrates • o ▪ Carbon dioxide ▪ Mineral salts ▪ Urea ▪ Creatinine ▪ Uric acid ▪ Excess water **Excretion in animals** • o ▪ Unicellular organisms ▪ Small animals ▪ Large animals ▪ Adaptations to habitat water - fresh and brackish e.g. marine desert terrestrial • o

LESSON NO.96 Excretory Organs ▪ Skin acts as excretory structure ▪ Salt glands are present in some marine or brackish water inhabiting organisms ▪ Intestine sometimes act as excretory organ ▪ Kidney o Major organ for excretion in vertebrates o Help also in osmoregulation Structure of kidney is different in organisms living in fresh and marine waters. Kidneys in freshwater organisms is modified to produce dilute urine while kidneys of marine or brackish water inhabiting animals are designed to produce concentrated urine.

External and internal structure of kidney Kidney is a bean shaped structure. A pair of kidneys is present in abdominal cavity attached to dorsal body wall. The concave part of kidney lies towards vertebral column. There is a depression present towards the vertebral column which is called hilus. The concave part provides space for entrance and exit of renal artery, vein and nerves. A thin tube ureter arises out of the concave part of kidney.

Renal cortex and renal medulla If we look at a section of a kidney, then the outermost layer, reddish in color is called cortex. After cortex the second layer consists of structure like pyramids occur. This region is called medulla. A funnel shaped cavity is present towards inside that receives urine called renal pelvis.

Nephron Nephron is the functional unit of kidney. Kidneys have many small structures present in cortex and medulla called nephrons. Function of the nephron is to produce urine. Nephron consists of following structures: o Glomerulus, which is a capillary network. o Bowman's capsule (Glomerular capsule) is the capsule like structure in which the glomerulus is present. o Convoluted tubule o Loop of Henle. Collecting duct

LESSON NO.97 Functioning of the kidney: Urine formation Function of the kidney is to produce urine. Kidney produces urine by following mechanism.

Pressure Filtration ▪ Pressure filtration in the renal corpuscle occurs due to blood pressure in renal artery. Molecules other than RBCs and plasma are filtered, i.e., these enter into the renal capsule. ▪ It is called Bowman's filtrate ▪ It consists of salts, glucose, urea and uric acid

Reabsorption The filtrate moves down the tubular system. Next step is called reabsorption. Reabsorption means that those compounds which are useful for the body return back to the blood and the remaining part of filtrate in the tubules makes urine consist of water and nitrogenous wastes. Reabsorption occurs through the network of capillaries surrounding the tubules. Reabsorption is selective process. Glucose, water and salts are reabsorbed. In the last step, only nitrogenous wastes remain inside the urine. This urine now goes down to pelvis and ureters. Through ureters it enters the urinary bladder for temporary storage. When bladder is full, there is a reflex for urination through urethra. Urination is not under conscious control in infants. In adults, however, this is under the conscious control through a sphincter present on the junction of urinary bladder and urethra.

LESSON NO.98 Osmoregulation Kidneys help in osmoregulation. This also helps in the maintenance of blood pressure. In case of increased blood pressure dilute urine is produced to temporarily reduce the quantity of water. In case of decreased blood pressure concentrated urine is produced and water is reabsorbed to compensate for fluid loss like in case of too much perspiration. Thirst is also a reflex in response to decreased quantity of water in blood.

LESSON NO.99 Ecology Ecology is the study of interactions among organisms and their environment.

Environment The physical surroundings of an organism including air, water, soil etc. Organisms have to interact with environment. Environment provides organisms everything required to live. Environment is always changing, may be supportive or hostile. Organisms living in an environment are "best fit" for that particular environment. Environment consist of biotic and abiotic factors ▪ Biotic factors Animals, plants, fungi, algae, bacteria ▪ Abiotic factors Air, water, soil What is Ecology? ▪ The study of interactions between organisms with each other and their environment. ▪ ▪ Biosphere and environment. Surface of Earth's crust Water bodies on Earth's surface • Atmosphere that surrounds Earth

LESSON NO.100 Levels of Organization in an Ecosystem Species ▪ Population ▪ Community ▪ Ecosystem ▪ Biome ▪ Biosphere

Ecosystem is the area where organisms live in interaction with non-living factors.

Organisms and their environment Organisms live in specific locations or areas suitable for them. Habitat is the term used for physical location of organism. Another term is niche. Niche is defined as ecological role of organisms. Organisms interact with their surroundings; their interaction is termed as niche.

Biotic and Abiotic Components ▪ Living / Biotic Components of an Environment Animals, plants; Interactions between organisms. ▪ Non-living / Abiotic Components of an Environment: Air, water; Interactions between organisms and their environment.

Energy Flow in Ecology Main source of energy in an ecosystem is "Sunlight". Autotrophs (producers) are the organisms which utilize the energy stored in inorganic compounds and use sunlight to make their food. These organisms carry out "photosynthesis" and use this energy to convert "carbon dioxide and water into oxygen and glucose". The other type of autotrophs makes carbohydrates using chemical energy e.g., many bacteria. These are called chemoautotroph. Heterotrophs (consumers) are the organisms that cannot make their own food and depends upon organic source. • o Herbivores (the organisms that eat plants, i.e., producers; also called primary consumers) o Carnivores (the organisms that eat herbivores or other carnivore animals) o Omnivores (the organisms that eat upon plants and animals both)

Pyramids in Ecology • o ▪ Decomposers (these are the organisms which break down the dead organic matter) ▪ Consumers (the organisms which feed upon producers or other consumers) ▪ Producers (the organisms which makes their own food by either using sunlight or inorganic chemical)

LESSON NO.101 Factors that Affect Environment • o ▪ Air currents ▪ Temperature ▪ Soil ▪ Water ▪ Light

Human interference: Industrial effluents Industrial growth results in the production of more and more effluents. Industries produce effluents which may be solid or liquid. These may contain heavy metals, toxic chemicals. These are disastrous for the biodiversity. These should be treated well. The solution is treatment plants for removal of dangerous materials.

Deforestation Forests are cut down for wood; wood may be used in buildings, furniture etc. Forests are also cut for making towns. This result in destruction of habitats of organisms which may leads to extinction of many organisms. As organisms are related, results may be increase or decrease of number of other organisms, e.g., if sparrows are decreased then insect pests may increase and destroy crops in that area.

Urban spread As populations are increasing there is an increasing need of towns and cities. Urban spread requires the cutting of forests or other type of habitat destruction.

Construction of dams, waterways Dams and water ways etc are constructed on water bodies. These may result in habitat destruction of fish and other aquatic animals. Loss of biodiversity results in instability of ecosystem.

Agriculture Agriculture also affect environment. In grasslands keeping too many animals may result into loss of grass and destruction of ecosystem. In crop growing fields fertilizer use and growing exhaustive crops (the crops that uses much of soil's resources) may result in damage to the ecosystem.

Natural factors Natural disasters like floods, earthquakes etc. may lead towards total or partial habitat destruction. Both biotic and abiotic factors are affected. The whole structure of ecosystem may change.

LESSON NO.102 Ecology II – Ecosystem :Ecosystems - An Introduction Ecosystem is an area where the organisms are living in an area in interactions with each other and their physical surroundings. There are many types of ecosystems, mainly divided into two categories: ▪ Aquatic ecosystems (ecosystem in water) ▪ Terrestrial ecosystems (ecosystem on land)

Basic components of an ecosystem; and their roles / significance

Ecosystem consists of: ▪ Abiotic factors Air, water, soil Temperature, pH, humidity ▪ Biotic factors Animals, plants, protists, bacteria

Biotic components • Producers(Plants, algae) • Consumers(Animals) • Decomposers(Fungi, bacteria)

LESSON NO.103 Pyramids in Ecosystems Ecological relationships are also shown in the form of pyramids in ecology. These are called ecological pyramids or trophic levels. On the base come producers, then consumers and then decomposers.

Producers The living organisms which make their own food are called producers. Examples include plants, bacteria and algae. They carry out photosynthesis, carbon dioxide and sunlight to water and oxygen. They synthesize carbohydrate by a process called Calvin's cycle. Entry point of energy in the ecosystem is producers.

Consumers These are the organisms that uses organic carbon source. These use carbohydrates produced by plants. Types include primary, secondary and tertiary consumers. Consumers include herbivores, carnivores and omnivores.

Decomposers These organisms feed upon dead organic matter. These include fungi and bacteria.

Abiotic components of Ecosystem :Water, Light, Air and Soil Water is a limiting factor in an ecosystem. It is required by living organisms. Organisms have to adapt according to the availability of water. • Sun light is the primary source of energy, e.g. zones of a lake. • Air manages temperature and other things. • Soil supports life

LESSON NO.104 ECOSYSTEMS – EXAMPLES Climate and weather affect the ecosystem. There are various ecosystems in the world and in Pakistan according to the climate of the area.

Aquatic Ecosystems Water is a liquid medium to support life. Aquatic ecosystems may be of fresh water or marine; a lake or a sea.

Water has some properties ▪ It changes its temperature slowly, more appropriate for life. ▪ It absorbs considerable amount of energy (sunlight) but at depths its level ▪ decreases for photosynthesis. ▪ Nutrients are concentrated at bottom. ▪ Water is abundantly available in this ecosystem.

Fresh water lakes ▪ o ▪ Lakes vary in nutrient, physical conditions and depths. ▪ Life also varies according to the conditions. ▪ Lakes have three zones: **Littoral** – shallow water, photosynthesis occur, most diverse, anchored plants, submerged plants, phyto and zooplankton and fishes. **Limnetic** – upper layer of deep water, have good light penetration, photosynthesis occur, cyanobacteria – protozoans – crustaceans and fishes.

Profundal – bottom layer of deep part, almost no light for photosynthesis, high nutrients on bottom, bacteria – decomposers.

Human interference Wastes comes from different areas and results in eutrophication. Excessive growth of cyanobacteria create a scum on the surface; result is that plants and animals die. Decomposer bacteria further decompose result in more organic matter. Ultimately habitat destruction is the result.

Terrestrial Ecosystems Plants and animals adapt to the changes from water to land habitat. Supporting tissue developed for the purpose; animals have skeleton and plants have vascular bundles to support these organisms on land. Water conservation is another adaptation to conserve water.

Types of Terrestrial Ecosystems Divided into four main types: ▪ Forest ecosystems o Tropical rain forests o Temperate deciduous forests o Coniferous alpine and boreal forests ▪ Grassland ecosystems ▪ Desert ecosystems ▪ Tundra ecosystems

o **LESSON NO.105 Major ecosystems in Pakistan** ▪ Pakistan has a variety of climate and seasons. ▪ Have these main ecosystems: o Temperate deciduous forests – Shogran and Neelum valley o Coniferous alpine and boreal forests – Kaghan, Malam Jabba, Dir and Chillas o Grassland ecosystems – Gilgit, Kashmir, Waziristan, Chitral and Kallat o Desert ecosystem – Mianwali, Bahawal Nagaar, Bahawalpur and more, Thal, Thar and Cholistan o Tundra ecosystem – mountains of Karakoram and Hindukash

Temperate deciduous forest ▪ o ▪ Present in Southeast Asia, India, North America, China, Japan etc. ▪ Have moderate temperature – 4 – 30°C ▪ Rain fall 750 to 1500 mm ▪ Plants – Taxus, Pinus, Berberis, ferns, grasses, herbacious; shed their leaves in dry season ▪ Animals – Rhesus monkeys, leopard, black bears ▪ Soil - very fertile and rich in nutrients ▪ Humans hunt animals and cut wood

Coniferous alpine and boreal forest ▪ Eurasia, North America, Canada ▪ Low temperature – freezing to 10C ▪ Snow cover is present ▪ Harsh climate – less suitable for life ▪ Highly adapted species lives here, Marco polo sheep, bison, wolf, black bear ▪ Plants – Pinus species grow; have long waxy needle like leaves to survive cold ▪ Human disturbs less because less accessible

Grassland ecosystem ▪ o ▪ Pakistan, Eurasian countries, North America ▪ Two types – Prairies, Savana ▪ Rain fall 250-750 mm, water is a crucial factor ▪ Plants are grasses – tall and short, legumes, herbs, mosses, lichens ▪ Animals are reptiles, amphibians, mammals ▪ Decomposers are fungi, bacteria ▪ Human impact – agriculture and livestock

Desert ecosystem ▪ o ▪ Rain fall – 25-50 mm ▪ Perennial plants, cacti, succulent leaves and stems ▪ Animals adapt to little water, kangaroo rats, reptiles, birds ▪ Human impact – desertification

Tundra ecosystem ▪ Very cold, snowy ▪ Small perennial flowers ▪ Mosquitoes grow well, birds feed upon these (geese, ducks) ▪ White bears, foxes, snow owls are present ▪ Human interference can produce long lasting effects because plants grow slowly but this interference is low due to harsh weather

LESSON NO.106 EVOLUTION Evolution is the branch of biology which deals with the study of change in the heritable characteristics of biological populations over successive generations. These characteristics are the expressions of gene, which are passed on from parents to offsprings during reproduction. Variations come from evolution and cause biodiversity. The history of evolution can be divided into 5 stages, classical, medieval, pre-darwinian, darwinian revolution and modern synthesis. In the classical era it was believed that all living things were related and that they had changed over time. One type of organism could descend from another type. Lucretius is one of the dominant philosophers of the classical era. In medieval period the concepts evolved from classical ideas, and it was believed that life forms were fixed. This concept is called fixed natural possibilities and states that all the life forms were fixed since the beginning of the time and would remain the same till the end. Aristotle added the concept that new types of living things could come to be.

Pre-darwinian period began in 17th century when philosopher and scientist began to explain natural phenomena in terms of physical laws. John Ray introduced the term 'species'. In 1735 Linnaeus explicitly recognized the hierarchical nature of species relationships and introduced the five-kingdom classification system. Buffon suggested that species could degenerate into different organisms. Erasmus Darwin proposed that all warm-blooded animals could have descended from a single microorganism. In 1809 Jean Baptiste Lamarck presented very first evolutionary scheme. He proposed the theory of changes/mutations over the time. It was a break from the concept of fixed creation, but his theory faced strong resistance.

Darwinian revolution is the next era in which Malthus's (1798) concepts about population growth and resultant struggle for resources. Seeking his inspiration from Malthus, Charles Darwin proposed the groundbreaking Theory of natural selection in 1859. Darwin's ideas were applied to humans by Huxley using paleontology and comparative anatomy. Gregor Mendel proposed the laws of genetics. Weismann for the first time made the important distinction between germ cells and somatic cells. He demonstrated that heredity passes through the germ line only. De Vries proposed that hereditary material is in the nucleus and transferred to cytoplasm of a cell.

The modern synthesis is the current age of evolution it connects natural selection and population genetics based on Mendelian inheritance, into a unified theory that applied generally to any branch of biology. Watson and Crick presented the structure of DNA. The study of evolution requires evidence based on biogeography, paleontology, comparative anatomy, and molecular biology. In this lesson the students will learn about • Evolution • Eras of evolution

LESSON NO.107 Theory of Natural Selection -I Based on the observations and the collected data, Darwin proposed his theory of natural selection in the form of 'Origin of Species' in 1859. This theory has four major postulates and each one describes the driving force behind the evolution. Competition; all organisms have a far greater reproductive potential than is ever realized. Female sea star releases about 1 million eggs each season, if all of them are fertilized and developed to reproductive adults by following year than it could result in a significant pressure on the resources on the habitat and can affect the sustainability of the ecosystem. The limited resources result in intraspecific and interspecific competitions between the individuals. In this lesson the students will learn about • Competition as the driving force behind evolution

LESSON NO.108 Theory of Natural Selection-II Individuals belonging to same population exhibit variations as the two individuals can't be same. Individuals vary in traits directly related to their ability to survive and reproduce. Inherited variations arise by random mutations, which can be either beneficial or disadvantageous.

These mutations could also be neither harmful nor helpful for the survival of the individual. Variations can be passed on to offspring. Individuals with beneficial or helpful variations can outperform the individuals lacking that variation. This results in increase of the population of the advantageous individuals. Mendel suggested that genes are responsible for traits, and these are inherited to offspring by transfer of genes. The phenomena of independent assortment and crossing over bring variations. Mendel's work confirmed that second postulate of the Darwin's theory. In this lesson the students will learn about • Variations as the driving force behind evolution

LESSON NO.109 Theory of Natural Selection-III Evolution as a result of competition and variation, leads to the individuals which are more competitive and capable of surviving at better scale in their habitat and niche. These individuals evolve in response limited resources and competition (both inter- & intra- specific). These individuals have higher rates of population growth and would increase their population. Reproduction provides variations and the traits which promote successful reproduction are said to be adaptive. This is the third postulate of the Darwin's theory, the survival of the fittest. Only those individuals/species will survive which have advantageous traits thus making them nature's favorite in evolution.

The last postulate of the theory of natural selection is the transfer of adaptive traits to next generation. Once an individual acquired any beneficial trait via reproduction, it will not only help the individual to survive but will also be transferred to the next generation. On the other hand, individuals with maladaptive traits are less competitive and are less likely to reproduce. With each generation, the population of individuals with maladaptive traits decrease and is eliminated eventually. This process of selection of individuals/traits is called natural selection. This selection is triggered by multiple factors affecting the environment (such as change in climate, food/water shortage, other stresses etc.), and will lead to extinction of ill-adapted species.

Alfred Russel Wallace is considered as the co-discoverer of the theory of natural selection. He collected data and specimens from Amazon and Malay Archipelago. Wallace independently proposed that populations out pressure on the resources of the habitat (competition) and variations arise in response to environmental changes. His theory has certain differences with the Darwin's natural selection. Wallace argued that each new variation arising on new generations is the result of adaptive radiations and are not necessarily caused by the changes in the environment. However, Darwin did not insist on finding adaptive significance for every modification. In this lesson the students will learn about • Survival of the fittest as the driving force behind evolution • Natural selection as the driving force behind evolution • Alfred Russel Wallace's theory of evolution

LESSON NO.110 FOOD CHAIN Ecosystem An area where the organisms are living in interactions with each other and their physical surroundings is called an ecosystem. There are many types of ecosystems, mainly divided into two categories: ▪ Aquatic ecosystems (water ecosystems) ▪ Terrestrial ecosystems (land ecosystems)

Ecosystem Ecosystem consist of biotic (living) and abiotic (non-living) components. ▪ Biotic factors: Producers, Consumers, decomposers Plants, animals, fungi / bacteria Biotic factors are in relationship with each other in an ecosystem. These also interact with abiotic factors present in their surroundings like air, water, sunlight.

Trophic levels Trophic levels or feeding levels in an ecosystem define roles of the groups of organisms. • Producers(Plants, algae) • Consumers(Animals) • Decomposers(Fungi, bacteria)

LESSON NO.111 Autotrophs – Producers These are the organisms that make their own food using sunlight as an energy source. These convert carbon dioxide and water into carbohydrates. Producers are the only source for all organic food in the planet Earth; this is the entry point of energy. These provide food to all other life forms. Examples include plants, algae.

Chemotrophs These are the organisms, which uses inorganic chemicals as energy source and makes carbohydrates. These lives in deep oceans, where there is no light source. Examples are some bacteria and deep-sea worms. **Heterotrophs – Consumers** Organisms that do not make their own food and get food from other organisms are called consumers. They get energy by eating either plants or other animals. Their source of food hence is organic compounds produced by producers. Examples are small and large animals.

Types of Consumers Consumers are divided into three classes: ▪ Primary consumers – herbivores (plant eaters) e.g., goats, cows ▪ Secondary consumers – primary carnivores (meat eaters that eat herbivores) e.g., frogs, snakes ▪ Tertiary consumers – secondary carnivores (meat eaters that eat other meat eaters) e.g., eagle, hawk

Decomposers These are the organisms that eat dead organic matter. These feed upon animals and plants – their fallen parts or dead organisms. These recycle food in the ecosystem, convert non available organic matter into available matter for producers.

LESSON NO.112 Food chains **What is a food chain?** Food chain is a one-way flow of energy in an ecosystem. Producers to consumers to decomposers In ecosystem different food chains exist. This is a relationship of eating and being eaten.

Phytoplankton à Fish Herbivore à Fish Carnivore à Fish Eater Bird: Example of a food chain from land (grassland) ecosystem **Grassland ecosystem** has following characteristics: ▪ Rain falls 250-750 mm, water is a crucial factor ▪ Plants are grasses – tall and short, legumes, herbs, mosses, lichens ▪ Animals are reptiles, amphibians, mammals ▪ Decomposers are fungi, bacteria Sunlight à Grass à Primary Consumer (Rat/Mole) à Secondary Consumer (Snake) à Tertiary Consumer (Hawk)

Example of a food chain from aquatic ecosystem ▪ Aquatic ecosystem is marked by: Water, which is a liquid medium to support life. May be fresh water or marine, a lake or a sea. ▪ Light, temperature penetration is important for biodiversity

A Simple Food Chain in a Lake Phytoplankton à Crustaceans à Fishes à Fishes

Arctic Ecosystem Algae à Shrimp à Arctic cod (fish) à Seal à Polar Bear

Food web What is a food web? ▪ A network of eating and being eaten. ▪ Feeding relationships between organisms are not as simple as food chains. ▪ There is a diversity of organisms, and one organism may be eaten by more than one other organisms. Example is grass is eaten by cows, buffaloes

LESSON NO.113 Example of a food web from land (grassland) ecosystem ▪ Grassland ecosystem have small plants, reptiles, amphibians and mammals. ▪ Grass is eaten by most of the herbivores. ▪ Herbivores are eaten by more than one carnivores.

Example of a food web from aquatic ecosystem Aquatic ecosystem has phytoplanktons, zooplanktons, plants, animals and decomposers. Food webs may be simple or complex starting from producers.

Desert Ecosystem ▪ Dry and hot environment. ▪ Few specific plants. ▪ Well adapted animals.

Forest Ecosystem ▪ Dense vegetation, animal diversity is high ▪ Simple to complex food webs

LESSON NO.114 ALGAE AND FUNGI Algae are: The organisms' range in size from microscopic unicellular to large sea weeds, demonstrate following characteristics: ▪ Major producers of oxygen ▪ Provide food for animals in various food chains ▪ Source of chemicals with pharmaceutical value ▪ Utilize carbon dioxide to reduce the greenhouse effect ▪ Source of valuable materials including biofuels, food and agar (part of media for microbial growth) ▪ Thrived for 1.5 billion years, are an integral part of the global ecosystem Distinguishing characteristics of algae, with examples ▪ Photosynthetic, autotrophic ▪ Eukaryotic, have cell wall consist of cellulose

- Classified by their:
 - o energy storage products
 - o cell walls
 - o colour (due to types and abundance of pigments their plastids)
 - o cellular organization
 - o Reproduce both asexually and sexually
 - o Asexually by mitosis
 - o Sexually when there is environmental stress
 - o Plus and minus gametes fuse to form zygospore which give rise to new organism e.g. Chlamydomonas
 - o Some reproduce by conjugation like spirogyra

LESSON NO.115 Types of Algae based upon the pigments ▪ Green algae (chlorophyll, soluble in organic solvents only, such as alcohol), 7000 species ▪ Red algae (have “phycobilins” water soluble pigment) 4000 species, cosmeics, gelatin etc ▪ Brown algae 1500 species, weeds ▪ Golden brown algae (formation of petroleum products) ▪ Carotenoids are also algal pigments soluble in organic solvents ▪ Euglenoids, both plant and animal like characteristics

Algae could be distinguished by cellular organization: ▪ Unicellular algae: species occur as single, unattached cells that may or may not be motile. For example, Chlamydomonas, Euglena. ▪ Filamentous algal species occur as chains of cells attached end to end. These filaments may be few to many cells long and may be unbranched or branched in various patterns. For example, Spirogyra. ▪ Colonial algae occur as groups of cells attached to each other in a nonfilamentous manner. For example, a colony may include several cells adhering to each other as a sphere, flat sheet, or other three-dimensional shape. For example, Volvox.

Distribution and Significance of algae Algae are widely distributed in various areas of the world in small to large water bodies. These are present in small ponds to huge oceans. Green algae are found almost everywhere. Red algae and brown algae are mostly marine organisms. Euglenoids are freshwater inhabitants. When algae grow a lot covering large area of water it is called algal blooms.

Important producers of the planet Earth Algae are producers in many food chains particularly in aquatic ecosystems. These convert sunlight into energy and carbon dioxide into carbohydrates. These help in reducing the Greenhouse effect by absorbing carbon dioxide.

Algae as human diet alternatives Agar is used in the place of gelatin to make gels in many foods. Weeds are part of the salads and other materials for making various kinds of salads and desserts. Some are used in making ice creams.

Production of Agar These are also used in production of Agar which is material extracted from sea weeds and help in solidifying bacterial growth media. ▪ Agar is very important in scientific research ▪ Used to culture microorganisms on solid surface ▪ A plentiful and cheap source for growing microorganisms ▪ Used in almost all microbiology research laboratories, for example in clinical laboratory

Algal Biofuels Algae can produce huge biomass. This biomass could be used to convert into fuels like ethanol. Algae are potential source of biofuels.

LESSON NO.116 The Fungi Fungi are the heterotrophs and major decomposers which feed upon decaying organic matter. These range in size from microscopic to large ones. These include heterotrophs, symbionts and parasites.

Distinguishing characteristics of fungi, with examples ▪ Unicellular to multicellular ▪ Eukaryotic ▪ Body is made of long filaments of hyphae (singular: hypha) which form a mycelium (collection of hyphae) ▪ Have a cell wall, made up of chitin ▪ Reproduce both asexually and sexually

- o Spores
- o Mating of hyphae

Some important groups of Fungi Phylum Zygomycota = the Bread Molds e.g. Rhizopus – black bread mold Phylum Ascomycota = the Sac Fungi e.g. Yeast Phylum Basidiomycota = the Club Fungi e.g. Mushrooms, puffballs, bracket fungi, rusts, smuts . Phylum Deuteromycota = the Fungi Imperfecti e.g. Penicillium

LESSON NO.117 Habitats and Significance of fungi Fungi grow in almost every habitat imaginable, as long as there is some type of organic matter present, and the environment is not too extreme. It is a diverse group; number of described species is somewhere between 69,000 to 100,000 (estimated 1.5 million species total). It

is present in soil, water, or on foods. These may also live on animal or plant bodies. Mycorrhizae are the fungi associated with the roots of vascular plants. Lichens are the association between fungi and algae.

Lichens Lichens consist of association between algae and fungi. It is a symbiotic association. These are usually the first inhabitants of a new ecosystem.

Fungi are important reducers / decomposers ▪ Feed on dead organic matter ▪ Convert it to less toxic form ▪ Convert into useable form for plant's use

Food source and food spoilage Many fungi are source of food and many other spoils foods. ▪ Mushrooms o Various types o Poisonous to edible o Used as food, protein source ▪ Some fungi cause spoilage of foods o Bread molds

Sources of medicines Some fungi are sources for medicines. ▪ Truffles o Used in medicines o Grow inside soil ▪ Penicillium o Source of penicillin ▪ Other drugs

Pathogenic fungi causing diseases in plants and humans Many fungi are pathogenic and cause infections in animals, plants and human beings. These may cause skin infections, eye infections and others. Parasites can harm human beings and plant infections can damage crops.

Yeast Yeast is a very important organism for biotechnology and food industry. It is used to make various biotechnological products. It is eukaryotic and better in making proteins in comparison to prokaryotes. It is also used in wastewater treatments because it absorbs heavy metals and converts other contaminants into less toxic form.

LESSON NO.118 ANIMALS AND THEIR IMPORTANCE

Distinguishing Characteristics of Animals, with examples ▪ Consist of eukaryotic cells ▪ Multicellular organisms, from simplest sponges to complex animals like mammals ▪ Highly adapted to their environments ▪ Widely distributed, present in all types of aquatic to terrestrial ecosystems even in air, e.g., the birds ▪ Important parts of food chains as consumers of different levels ▪ Heterotrophic, rely on organic carbon sources ▪ Independently living or parasitic ▪ Reproduce asexually or sexually ▪ Very important for human beings and the ecosystems

Diversity of animals These are widely distributed and are present in almost all habitats. There are two major groups including invertebrates and vertebrates. Animals are wild if these are living in natural habitat, or these are termed as domestic if these are kept for some benefit or as pets. Animals are kept in captivity or controlled habitats if there is a need in zoological gardens and safari parks.

LESSON NO.119 Phyla of Invertebrates with examples: Porifera These are simple animals with one opening as mouth and a body cavity. These are aquatic organisms. ▪ e.g. Sponges ▪ Two layered, no specific organ, porous body ▪ Highly specialized cells

Coelenterates The animals with 3 body layers. These have stinging cells which have a poison to paralyze or kill the prey. ▪ e.g. Jelly fishes which are marine ▪ Have simple organs ▪ Have stings, mouths, tentacles

The worms These are of different kinds including round worms, flat worms and earth worms. ▪ Regeneration: These have ability to regenerate their lost part. ▪ Parasites, decomposers ▪ Grow very long

Arthropods, widely distributed These are one of the most diverse groups of animals. These organisms have an exoskeleton. ▪ Four major groups o Arachnids (spiders) o Crustaceans o Insects o Millipede, centipede ▪ Have sense organs, are segmented, have exoskeleton

Molluska These organisms are specific to have shells around their soft bodies. ▪ Soft bodies, hard shells ▪ Aquatic mostly ▪ Snails, lobsters ▪ Some of these make pearls

Echinoderms These are the animals with spiny skins hence called echinoderms (echino-spiny). ▪ Exclusively marine ▪ Have a water vascular system for the movement ▪ Examples are star fish, sea cucumber, brittle star

Importance of Invertebrates ▪ Invertebrates are very important part of lot of food chains hence important for stability of ecosystems ▪ For human use: o Sponges are used widely in sound proofing, washing o Worms are important parasites of domestic animals and human beings o Insects are pests of many crops, many useful insects like honeybee, lac insect o Lobsters makes pearls, cultured for pearls

These are components of Food Webs ▪ Component of aquatic food webs ▪ Components of terrestrial food webs ▪ Maintain ecosystem stability, if one becomes extinct then the whole food chain or web may become disturbed.

Role in Soil Fertility ▪ Some invertebrates are decomposers, e.g., Earthworm ▪ They feed upon dead organic matter and increase soil fertility ▪ When animals die their bodies become part of soil and decomposed to add nutrients to soil

Parasitic Invertebrates These could harm the humans and also harm our domestic animals and crops. There are many treatments available for these. ▪ Ticks, mites, lice are ectoparasites ▪ Worms are endoparasites ▪ Leeches are ectoparasites

LESSON NO.120 Major Classes of Vertebrates and their Importance Vertebrates are the organisms with a vertebral column precisely with 3 main characteristics at any stage of their life including notochord, nerve cord and pharyngeal slits. Vertebrates are divided into: ▪ Fishes ▪ Amphibians ▪ Reptiles ▪ Birds ▪ Mammals

Fishes ▪ Aquatic animals, occupies fresh water and marine habitats, widely distributed, bony or cartilaginous ▪ Have gills for gas exchange, 5 chambered single circuit heart ▪ Have streamlined and slimy body, ectotherms ▪ Have fins (appendages) and tail ▪ Cultured in ponds and caught from natural habitats, are important food source of proteins ▪ Cultured for ornamental purposes ▪ Examples are Rohu, Carps, sharks, electric rays

Amphibians ▪ Are transition between aquatic and land animals ▪ Gills, lungs and cutaneous respiration ▪ 5 chambered double circuit heart, mixing of blood ▪ Some part of the life is aquatic and some part terrestrial ▪ Produce large number of eggs, no parental care ▪ Ectotherms ▪ Examples are frogs, toads

Reptiles ▪ Of diverse types, widely distributed, aquatic and marine ▪ Highly adapted for hot dry habitats such as deserts ▪ Terrestrial animals, ectotherms ▪ Scaly skin, four chambered double circuit heart ▪ Lungs for gas exchange ▪ e.g. lizards, chameleon, turtles, tortoises, crocodiles, snakes, gavials

LESSON NO.121 Birds ▪ Aerial mode of life, terrestrial life, fore arms modified into wings ▪ Endotherms, 4 chambered heart ▪ Lungs and air sacs for respiration ▪ Lays fewer eggs, provide parental care ▪ Bones light weight called hollow bones ▪ Grain eaters to omnivores ▪ e.g. parrots, pigeons, peacock, pelican, king fisher

Mammals ▪ Widely distributed, most complex, endotherms ▪ 4 chambered heart, lungs for gas exchange ▪ Highly developed brains ▪ Young ones are born mostly, mother feed them with milk, provide parental care ▪ Have hairs on their body, provide insulation ▪ Thick skin with skin sensory receptors ▪ Heterotrophs, herbivores to omnivores

Importance of Vertebrates These are beneficial ▪ Important parts of food chain and webs ▪ Beauty and diversity of nature ▪ Provides protein foods to human beings, eggs, milk, meat, honey ▪ Furs and feathers are used for various purposes ▪ By products of fish industry ▪ Pearls, clam culture ▪ Lac insect and silk worm ▪ Snake venom for medicines

These are harmful ▪ Poisonous animals, e.g. frogs, snakes, ▪ Parasitic animals, e.g. worms, mites, ticks ▪ Insect pests of crops ▪ Other animal pests e.g. rats ▪ Termites damage woody structures of buildings

LESSON NO.122 REPRODUCTIVE SYSTEM **Introduction of reproduction** Reproduction is one of the basic characteristics of life. It is a biological process by which the organisms produce their young ones which are similar to their parents. This ability also allows organisms to adapt itself to the changing environment.

Types of Reproduction There are two types of reproduction, asexual and sexual. In asexual reproduction, offspring are produced by simple cell division. No gametes are involved, and offspring have same characteristics as their parents. In sexual reproduction, two organisms are involved in reproduction and offspring have characteristics which are combination of their parents and some new. Gametes are involved in this kind of reproduction. Sexual reproduction has advantage of increase in genetic variability. Gametes are formed by a process called meiosis in which alterations in genetic material takes place. As two organisms are involved, mixing of their characteristics takes place that increase more genetic variability. Genetic variability leads towards more adaptability to the environment.

LESSON NO.123 Types of asexual reproduction Asexual reproduction involves no gametes. There are many methods of asexual reproduction in organisms including: ▪ Binary fission ▪ Multiple fission ▪ Budding ▪ Regeneration ▪ Vegetative propagation in plants

Binary fission This is a type of asexual reproduction in which one unicellular organism divides into two by simple division. In bacteria, for example, binary fission takes place: 1. Replication of single chromosome takes place. 2. Daughter chromosomes move towards opposite poles. 3. Cell membrane invaginates and meets in the center. 4. Cell wall follows the cell membrane and cell divides into two. Binary fission also takes place in unicellular organisms like amoeba and paramecium. In these organisms: 1. nucleus elongates and divide into two 2. Cytoplasm also divides and cell divides into two

Multiple fission This is a process in which a single cell divides into many cells. For example, amoeba makes a cyst in unfavorable conditions. Its nucleus divides into many nuclei and every part is surrounded by some cytoplasm. When favorable conditions come, the cyst split into many cells, each one a new amoeba.

Budding In this process an outgrowth (called bud) appears on the surface of an animal and may separate later on to develop into a new organism. Examples are yeast, hydra.

Regeneration Regeneration is the formation of lost part of the body. Examples are starfish and planaria. If their one body part is lost then they have ability to grow it again. In star fish an arm is lost then the arm grows again.

Vegetative propagation in plants It is a process that involves the vegetative propagation of some plant part such as stem or leaves. Examples are runners, suckers. Runners like grasses and strawberry and suckers like banana and mint.

LESSON NO.124 Sexual reproduction in Plants Flowering plants reproduce by sexual reproduction. Flowers have male and female parts called stamen and carpel. Fertilization occurs with the help of a process called pollination, i.e., transfer of pollen grains from anthers of stamen to carpel and then ovary. Pollination is of two types: self-pollination and cross pollination. Formation of seed and fruit ▪ Seed is formed from the zygote. It consists of embryo and its food. ▪ When the seed are formed then the wall of ovary becomes fleshy or scaly and forms fruit wall. ▪ The floral parts (sepals, petals etc.) fall off. ▪ The ripened ovary is called fruit.

LESSON NO.125 Sexual reproduction in Animals In animals, it occurs through fertilization of gametes (sperms and ova). Gametes are produced by meiosis which is the reduction division. These have half a number of chromosomes. Females and males are different in only one pair of chromosomes. Males have a chromosome different called a Y chromosome in one pair. So, females have an XX pair while males have XY.

Example of Frogs ▪ Frogs carry out sexual reproduction. ▪ Female frog lays a large amount of ova in water and male release sperms on these. ▪ Sperms and ova fertilize to produce zygote. ▪ Zygote the divide by mitosis to develop a larva which lives in water for a specific period of time and then by a process called morphogenesis develops into adult frog.

Reproductive system in humans Humans reproduce sexually by fertilization of gametes called sperms and ova. Gametes are produced by specialized organs called testes and ovaries. Gametes have a haploid number of chromosomes, i.e., 23 chromosomes. 22 of these pairs are alike but 23rd pair is different in males and females. Females have X chromosomes in their gamete while males have an X or Y. Gender of the child is hence determined by the sperm, X or Y. Female always contribute an X in gamete. If male gamete has a Y then a male child will be produced. If sperm have an X then female child will be produced.

Basic components of male reproductive system Human male reproductive system consists of a pair of gonads called testes, and a system of accessory tubes. (testis: sing.; testes: plural). Testes are paired which produces sperms. Testes lie outside the body in the sac called scrotum. From testes, sperms are transferred to main duct called vas deferens. Vas deferens makes a complex system of ducts called epididymis. From epididymis, sperms pass through the urinogenital tract and discharged.

Function of testes Function of testes is to produce sperms and the male hormone testosterone. Testis consists of a system of tubules called seminiferous tubules.

Process of Spermatogenesis 1. In the seminiferous tubules, germinal epithelium produces the cells called spermatogonia. 2. Spermatogonia divide into primary spermatocytes (2N). 3. Primary spermatocytes undergo meiosis to produce secondary spermatocytes (N). 4. Secondary spermatocytes divide further to make 4 haploid cells called spermatids. 5. Spermatids are converted to sperms by morphological changes. Testis also has some large cells in seminiferous tubules called sertoli cells. These cells provide support and nourishment to the developing sperms.

Basic components of female reproductive system Female reproductive system consists of a pair of ovaries and an associated system of tubes. A pair of ovaries lies inside the female body cavity which produces ova. Ovaries leads to the tubes called oviducts also called fallopian tubes. Oviduct opens into uterus. Ovum is fertilized in the oviducts then it passes to the uterus where it is implanted. A placenta is established between the uterine and fetal tissues for the exchange of materials. Uterus is connected to the vagina through cervix.

Function of ovaries Function of ovaries is to produce ova and female hormones estrogen and progesterone. Germ cells in ovary produce many cells called oogonia.

Process of Oogenesis 1. Oogonia produce primary oocyte (2N) by mitosis. 2. The primary oocyte divides by meiosis into secondary oocyte (N) and polar body. 3. The secondary oocyte divides to form one ovum and a polar body. 4. The first polar body also divides to form 2 polar bodies. 5. Hence from secondary oocyte, one ovum and three polar bodies are produced. Usually, ovary releases one ovum at a time, this is called ovulation. Sometimes it releases more which may result into fraternal (non-identical) twins.

LESSON NO.126 APPLICATIONS OF BIOLOGY Biology – Useful for Humankind and World ▪ Demands and problems are increasing with increase in population. ▪ Biologists investigate phenomena of nature and problem and find ways to resolve these. ▪ They introduced techniques and technologies to explore all biological resources to cope demands and resolve problems.

Biotechnology ▪ Biotechnology is the use and manipulation of biological processes, organisms and systems to manufacture products to improve the quality of human life. ▪ The term biotechnology was first used in 1970's. ▪ Some important applications of Genetic Engineering of Fermentation and Microorganisms

Genetic Engineering ▪ It is a process in which a useful gene is taken from one organism and inserted into another organism. The resulting organism is called transgenic. ▪ This process is carried out with the help of vectors and enzymes. ▪ Most common vector is bacterial plasmid. ▪ Most common enzymes are restriction endonucleases or restriction enzymes.

LESSON NO.127 Transgenic bacteria ▪ E. coli – most common microorganisms in biotechnology because these have many strains (types) which are non-pathogenic, i.e., these cannot produce disease. ▪ GRAS (Generally regarded as Safe) Organisms are those which are known for their non-pathogenicity. ▪ Chemical industry: Acids and alcohols are produced using bacteria. ▪ Pharmaceutical industry o Formation of insulin by E. coli for diabetic patients o Formation of human growth hormone for dwarf children o Formation of vaccines

Transgenic Bacteria Transgenic bacteria are used in many industries, for example: ▪ Food industry o Yogurt, Cheese, Sour creams are prepared using microorganisms o Breads b use of yeast o Beverages, drinks by yeasts or some bacteria ▪ Treatment of waste materials is also carried out using different form of bacteria or even yeast. The microorganisms are used to treat: o Solid wastes o Wastewater o Oil spills o Nuclear wastes ▪ Bacteria and algae are also helpful in biofuels production (biogas, biodiesel).

Transgenic plants These plants are produced to develop certain specific properties in these, for example, resistance to viruses. Some examples are following: ▪ Virus resistant crops (cotton) ▪ Insect resistant corn ▪ Genetically modified banana ▪ Pesticide resistant crops ▪ **FOOD SAFETY AND SECURITY:** This is very important in modifying food crops particularly that if we are inserting a gene of resistance or else, the plant should not develop any harmful characteristic. For this purpose, before selling any transgenic seed or other product to market, it is tested for safety. This is called biosafety.

Transgenic animals ▪ Transgenic mice as model of human disease o Obese rats, diabetic mice ▪ Genetically modified organisms to increase milk yield ▪ Genetically modified farm animals to produce antibodies ▪ **FOOD SAFETY AND SECURITY:** This is very important in modifying food producing animals particularly that if we are inserting a gene of interest the animal should not develop any harmful characteristic. For this purpose, before selling any transgenic product to market, it is tested for safety. This is called biosafety.

Fermentation Pathways Microorganisms' carryout fermentation, i.e., utilization of anaerobic pathway for production of energy. They produce many useful products for us during these pathways. These products are commercially produced by growing such organism in mass culture in very big vessels called fermenters or bioreactors. Following products are obtained by this process: ▪ Biofuels ▪ Acids and alcohols ▪ Yogurt, cheese, sour creams ▪ Bread ▪ Beer, wine

LESSON NO.128 Applications of biology in culturing Animal: Animals are traditionally cultured for their products ▪ Meat from fish, chicken, goats, cows ▪ Milk from cows, buffaloes ▪ Eggs from hens, ducks ▪ Wool from sheep ▪ Leather industry uses skins ▪ Honey from Honeybee ▪ Silk from silkworm

Common types of animal cultures ▪ Aquaculture ▪ Ornamental fish ▪ Seri culture (Silkworm) ▪ Dairy and Poultry ▪ Apiculture (honeybee) ▪ Biological pest control We discuss these one by one briefly.

Aquaculture Culturing animals in water is called aquaculture for example fish farming. ▪ Fish farming, Prawn culture ▪ Pearl oyster culture ▪ Other aquacultures (clams, lobsters etc.) ▪ Byproducts of fishing industry include many cosmetics and some animal diets. Biology helps in improvement of fish varieties and also extraction of various products from these, for example: ▪ Selection of fishes in a fishpond (herbivores, carnivores) ▪ Feed for fishes which may increase their production ▪ Diseases and their treatment: biology help in finding causes of diseases and their treatments.

Aquaculture for Ornamental Fish Ornamental fishes are the beautiful usually small fishes kept in small aquaria in houses or other places like hotels. Biology helps in: ▪ Feed: which is good for their maintenance ▪ Diseases: how can we handle diseases ▪ Requirements for healthy culture in small to large aquaria

Sericulture Culture of silk moth for making silk on mulberry leaves is called sericulture. It is a popular home industry in many parts of the world Biology helps in improvement of this culture: ▪ Process of making silk ▪ Diseases in moth or

larvae can result in huge loss ▪ Biology helps in making new varieties, e.g., wingless variety, varieties which completes 5-6 cycles in a year

LESSON NO.129 Dairy and poultry Biology plays very important roles in dairy and poultry industries. Biology helps in many ways for example: ▪ Improved breeding techniques and cross breeds (more vigorous and resistant). ▪ Improved quality and quantity yield of meat, milk, wool and eggs. ▪ Diagnosis and cure for diseases in farm animals. ▪ Biological knowledge is also helping in providing balanced food for best yield.

Apiculture (honeybee) Keeping the honeybee for honey production is called Apiculture (Bee keeping). Biologists explore the phenomena behind Honey production, like knowledge of biology tells us that: ▪ Honeybee is a social insect and lives in large colonies ▪ Queen bee, drones and workers Biology helps in understanding how this system works and can help in production of better varieties and improved system for culturing these animals to increase yield of honey.

Biological pest control A large number of organisms damage the useful plants or animals and can cause huge losses. These are called pests, e.g., many insects damage crops.

There are few ways to control pests: ▪ Primary or cultural control o Crop rotation rather than monoculture, i.e., different crops are grown at different seasons. ▪ Chemical control o Various chemicals are used to kill insects but insects may develop resistance and chemicals are dangerous for health ▪ Biological control means that the pests are controlled or killed by living organisms. o This is the most promising method of pest control o The natural enemies and predators of the pests are used to kill the pests o Lady beetles for citrus fruits o *Bacillus thuringiensis* for cotton

Selection of biological control agent needs care. These should be ecologically safe. For example if we introduce a sparrow for eating an insect and that sparrow also start eating some plants. This will be even more harmful for crops and environment.

Advantages of biological pest control ▪ They do not pollute the environment. ▪ They do not disturb the natural ecosystem. ▪ A more permanent control because pest usually do not develop resistance

Applications of Biology in agriculture and plant growth: Plants are in many ways useful ▪ Parts of ecosystems, producers ▪ Food crops: crops, which makes our food like wheat, rice, vegetables. ▪ Many useful products like perfumes, medicines, furniture ▪ Beauty of the nature, ornamental plants

Biology helps in: ▪ Understanding the life cycles and methods of spread ▪ Better varieties by cross breeding or genetic engineering ▪ Disease mechanisms and control ▪ Appropriate soil, fertilizers and water availability

LESSON NO.130 Science, Ethics & Values Science: Latin term “Scio” Observation and theoretical form Observation and experiment

Science: ▪ Investigation of the universe by a set of methodologies ▪ Progress made by scientific methods ▪ Stepwise, not a single activity, not a value free

Ethics: ▪ Associated with science ▪ Issues arises from scientific research ▪ Scientists are trying to do so

Values: ▪ Science has entered into our daily lives ▪ Proper resource allocation ▪ Reflects what society at the time deems to be valuable.



(I have prepared these concise notes specifically for printing and quick revision. The original handouts are often quite lengthy, making revision difficult and printing expensive. To solve this, I have summarized each lesson into fewer pages to ensure a cost-effective and efficient study experience.

The BIO101 notes is same as official handouts but in less pages (without diagrams), while the remaining subjects have been compiled using AI notebooks. My primary goal was to create a resource that allows for quick revision without being heavy on the pocket.

I am sharing these in the group so that others may also benefit from them. I am a student of BS Accounting & Finance. If you have any study groups for group study or academic assistance of accounting, please add me: 0318-2029650)